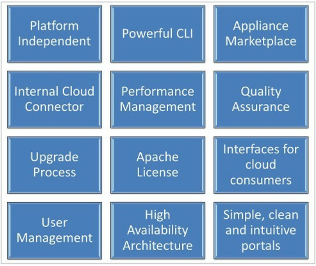

Figure 1: Key features of OpenNebula

directory service through which you can add new users, and those users can be individually entitled. Managing systems, configuring new virtual systems or even targeting the right users and groups is easy in OpenNebula.

- ***Availability at all times***

OpenNebula not only takes care of the initial provisioning, but the high availability of its cloud environment is much better compared to other cloud solutions. Of course, the central OpenNebula services can be configured for high availability, but this is not absolutely necessary. All systems continue to operate in their original condition and are automatically included in the restored availability of the control processes.

- ***Easy remote access***

In virtual environments, one lacks the ability to directly access the system when there are operational problems or issues with the device. Here, OpenNebula offers an easy solution — using the browser, one can access the system console of the host system with a VNC integrated server.

- ***Full control and monitoring***

All host and guest systems are constantly monitored in OpenNebula, which keeps the host and VM dashboards up to date at all times. Depending on the configuration, a virtual machine is to be restarted in case of the host system failing or if migrating to a different system. If a data store is used with parallel access, the systems can of course be moved, while in operation, on to other hardware. The maintenance window can be minimised and can often be completely avoided.

- ***Open standards***

OpenNebula is 100 per cent open source under the Apache License. By supporting open standards such as OCCI and a host of other open architecture, OpenNebula provides the security, scalability and freedom of a reliable cloud solution without vendor lock-in, which involves considerable support and follow-up costs.

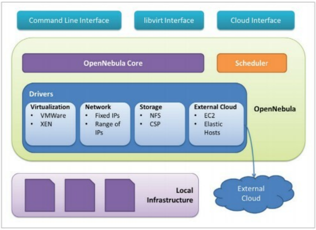

Figure 2: OpenNebula architecture

OpenNebula architecture

To control a VM's life cycle, the OpenNebula core coordinates with the following three areas of management:

1) Image and storage technologies — to prepare disk images
2) The network fabric — to provide the virtual network environment
3) Hypervisors — to create and control VMs

Through pluggable drivers, the OpenNebula core can perform the above operations. It also supports the deployment of services.

VM placement decisions are taken by a separate scheduler component. It follows the rank scheduling policy, which makes place for VMs on a physical host according to the rank given by the scheduler. These ranks are decided by the scheduler, using a ranking algorithm.

OpenNebula uses cloud drivers to interact with external clouds, and also integrates the core with other management tools by using management interfaces.

Components of OpenNebula

Based on the existing infrastructure, OpenNebula provides various services and resources. You can view the components in Figure 3.

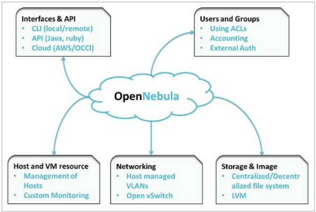

Figure 3: OpenNebula components